Adaptive Learning Tool to Enhance Educational Outcomes for Students with Inattentive Attention Deficit Hyperactivity Disorder (ADHD)

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***Abstract*—This paper presents an adaptive learning tool to enhance educational outcomes for students with inattentive ADHD. The platform comprises four key components: content chunking and simplification with interactive content, an adaptive quizzing system, a personalized ADHD identifier and break recommendation system, and a tool to maximize learning potential. The content chunking component transforms traditional educational materials into ADHD-friendly formats using Natural Language Processing (NLP) technologies, making them more accessible and engaging. The adaptive quizzing system employs a modified spaced repetition algorithm to optimize memory retention and engagement through personalized quizzes. The personalized ADHD identifier uses a hybrid Convolutional Neural Network (CNN) model to analyze webcam data, tracking head movements and facial emotions to provide real-time break recommendations, enhancing focus and cognitive performance. The final component integrates machine learning and data mining to adaptively assign tasks and provide personalized feedback, fostering a supportive learning environment. Preliminary evaluations indicate significant improvements in academic performance and engagement among ADHD-diagnosed students. This research underscores the potential of adaptive learning technologies in creating inclusive educational environments addressing the unique challenges faced by students with ADHD.**

***Keywords*—ADHD, adaptive learning, personalized education, memory retention, engagement detection**

I. Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder affecting approximately 5-6% of adults and 8-10% of children worldwide [1]. The primary symptoms of ADHD, including difficulties with attention, hyperactivity, and impulsivity, pose significant challenges in traditional educational settings. These environments often fail to meet the unique needs of students with ADHD, resulting in poor academic performance, low engagement, and increased dropout rates.

Students with ADHD struggle to maintain focus, organize tasks, and retain information-key skills for academic success. Traditional educational methods frequently do not address these challenges, creating a mismatch between teaching approaches and the learning needs of these students. This underscores the need for innovative educational tools that adapt to the cognitive and behavioral characteristics of students with ADHD. This research develops a tailored educational platform for students with ADHD, integrating advanced technologies to provide personalized learning experiences that enhance engagement, improve memory retention, and support better academic outcomes. It employs adaptive learning technologies, machine learning, and data mining to create a supportive and effective learning environment.

Adaptive learning technologies have shown significant potential in transforming education by personalizing the learning experience. These technologies dynamically adjust content and instructional strategies based on individual student performance and engagement levels. For students with ADHD, such personalized approaches are particularly beneficial as they provide immediate feedback, tailor difficulty levels, and maintain high levels of engagement through interactive and varied content. Existing research underscores the importance of frequent and adaptive assessments in improving academic achievement and memory recall. For students with ADHD, adaptive quizzes that provide immediate feedback and adjust difficulty levels based on performance can significantly enhance engagement and retention [2]. Personalized learning tools that account for individual strengths and weaknesses can create a more inclusive and effective educational environment [3],[4].

This study focuses on developing and evaluating an adaptive learning platform designed specifically for students with inattentive ADHD. The platform aims to transform generic educational materials into ADHD friendly formats, improve knowledge retention through adaptive quizzing, and provide personalized support and recommendations based on real-time behavioral and cognitive data. The proposed

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platform seeks to bridge the gap between traditional teaching methods and the learning needs of students with ADHD, enhancing educational outcomes and fostering a more inclusive learning environment. This innovative approach aligns with the goals of technology-enhanced learning, offering scalable and effective solutions for diverse educational contexts [5].

II. Literature Review

Students with ADHD face significant obstacles in traditional educational settings due to inattention, hyperactivity, and impulsivity, which hinder their academic performance and concentration [6]. Research indicates these students often struggle with study skills, note-taking, summarizing, and test-taking [6]. Effective interventions often involve structured information, personalized learning experiences, and frequent feedback [7]. Technology, such as adaptive learning tools and assistive applications such as text-to-speech and speech-to-text, has been shown to improve academic performance [8]. Moreover, the integration of Universal Design for Learning (UDL) principles ensures that learning materials are accessible and engaging for all students, including those with ADHD [9].

Natural Language Processing (NLP) technologies offer powerful tools for analyzing and transforming educational content. NLP can be used to automatically identify and highlight key points, generate summaries, and restructure content, significantly enhancing learning experiences for students with ADHD [10].

Adaptive quizzing tools tailored for students with ADHD enhance memory retention, engagement, and motivation by offering frequent and personalized assessments. These tools significantly improve academic performance. Weekly multiple-choice quizzes (MCQs) have shown substantial improvements in student performance, highlighting the positive impact of frequent quizzing on learning [11].

Frequent quizzing encourages thorough preparation and active class participation. Immediate feedback helps students identify strengths and weaknesses, enhancing the learning process. Regular quizzes establish consistent study schedules, reinforcing learning and ensuring better information retention. Additionally, frequent quizzing serves as a predictive tool for student outcomes, offering opportunities for early intervention to support at-risk students [12], [13]. Despite these positive findings, gaps remain in understanding the long-term impacts of frequent quizzing on diverse student populations, including those with ADHD. Future research should focus on longitudinal studies to evaluate the sustained benefits of adaptive quizzing tools and investigate optimal quiz frequency and timing to maximize effectiveness [14].

Multiple studies have explored the use of technology for ADHD detection. One study demonstrated the efficacy of using eye movement data recorded via webcams to detect ADHD in adults, achieving an accuracy of 82% using a logistic regression classifier [15]. Another study developed a rule-based system that utilized eye movement data to identify ADHD, emphasizing the potential of non-intrusive, cost-effective methods for ADHD screening [15]. These studies highlight the potential of using machine learning and computing vision for ADHD detection. However, research gaps remain as most studies focus on either ADHD detection or engagement monitoring separately, without integrating these into a comprehensive system. Furthermore, there is limited research based on real-time data to support students with ADHD in e-learning environments [15].

This research addresses these gaps by developing a holistic system that combines ADHD detection, engagement monitoring, and personalized break recommendations. This system identifies ADHD and determines the ADHD level, incorporating cognitive level as a factor. By integrating these components, this research aims to provide a scalable, efficient solution that enhances learning outcomes for students with ADHD [17].

Engagement detection in e-learning environments has gained attention to improve learning outcomes. One study proposed a hybrid CNN architecture to monitor student engagement through facial expressions, eye gaze, and head movements, achieving an accuracy of 88% [16]. Combining facial recognition with engagement detection models significantly improves the accuracy of identifying engaged and disengaged states during online learning sessions [17]. Educators and researchers face challenges in improving outcomes for ADHD students. The use of adaptive learning tools has become increasingly important to address their diverse educational needs. Learning Management Systems (LMS) data predict performance across various metrics by analyzing interactions with online materials, such as assignment submissions [2]. Machine learning algorithms using internal assessments, demographic data, and academic history help educators by offering valuable insights to identify at-risk students and tailor interventions [4]. Studies show consistently lower achievement, emphasizing the need for self-monitoring techniques [18]. CollabCrew, an automated task allocation system using machine learning, optimizes productivity by dynamically adjusting tasks based on past performance and skills [3]. The personalized learning experiences tailored to individual styles evaluate an adaptive learning system in online courses [19]. The system presents materials based on preferences, enhancing exam performance and overall learning outcomes [18].

III. Methodology

This research addresses four critical components designed to support students with ADHD: Content Chunking and Simplification using Interactive Content, an Adaptive Quizzing System to improve memory retention, a Personalized ADHD Identifier with a Break Recommendation System, and strategies to maximize learning potential among undergraduate students with ADHD. Fig. 1 provides a comprehensive system overview, illustrating how each component enhances focus, memory, and engagement in academic settings. By fostering individualized support and adaptive learning, this framework promotes sustained attention and retention of educational material tailored to ADHD students. The interactive and adaptive approach highlights the potential for creating learning environments that accommodate diverse cognitive needs, promoting an inclusive and effective educational experience.

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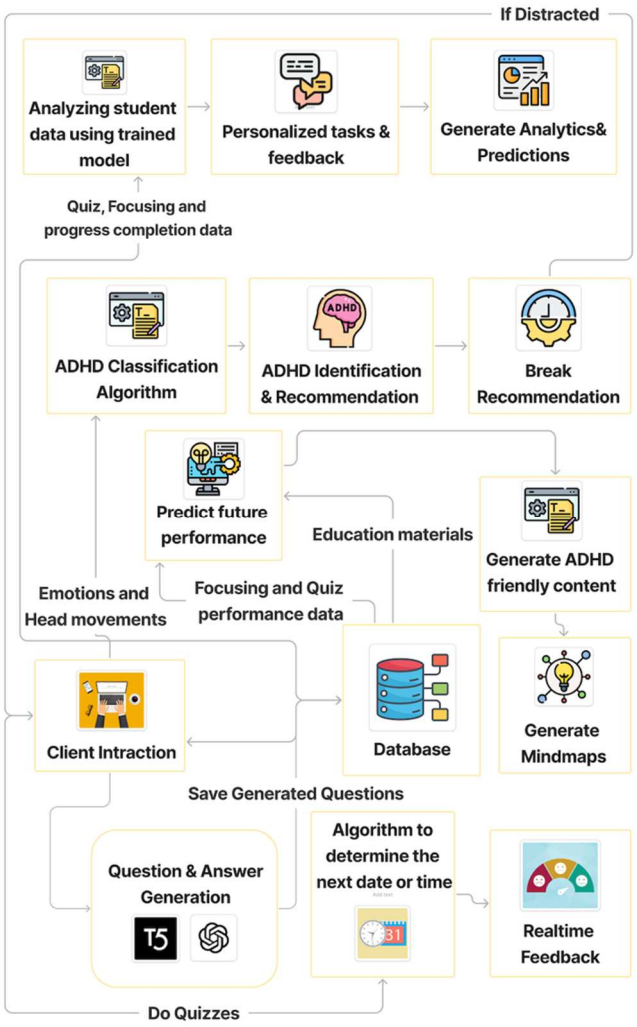


Fig. 1. System Overview Diagram

To understand the content needs of students with ADHD, a study was conducted involving interviews with over 10 ADHD individuals and consultations with psychiatrists and therapists at Peradeniya and Kandy Hospitals, supplemented by an online survey on students' learning preferences. The goal is to transform traditional educational materials into ADHD-friendly formats using a structured content transformation process. This process includes sequential stages: content ingestion, preprocessing, generation, and visualization. Initially, educational materials are extracted with Python libraries and preprocessed with NLP tools such as NLTK and spaCy, which streamline text by removing irrelevant details, tokenizing, stemming, and lemmatizing, thus reducing cognitive load. A customized T5 NLP model simplifies this content into structured formats such as bullet points, while GPT-4 enhances clarity with additional examples where needed. The simplified content is then organized into interactive mind maps created using NetworkX and D3.js, allowing students to explore concepts flexibly and non-linearly, supporting self-paced learning. Customizable presentation features, such as font size and color schemes, further enhance accessibility. The system continuously adapts content based on real-time feedback and engagement data, tailoring presentations to individual needs and progress, and creating a responsive, ADHD-friendly learning environment.

To assess the effectiveness of the adaptive quizzing tool, a quantitative experimental approach was adopted. Quizzes were generated by a fine-tuned T5 transformer model, trained on the SQuAD dataset to ensure high-quality question generation, and tokenized using the SentencePiece library for optimized language processing. Phrases and sentences were extracted as potential answers through GPT-4-mini, with three distractors created for each quiz item to enhance assessment accuracy and challenge users.

The adaptive review scheduling was guided by an adapted SuperMemo-2 (SM-2) algorithm, which calculates optimal review intervals based on individual response patterns.

The interval I' is updated using Equation (1):

$$I' = \min(\max([M_R \times I], 1), D_{max}) \quad (1)$$

Where M_R represents the response-dependent multiplier, and D_{max} is the maximum allowable interval.

The ease factor EF' is adjusted using Equation (2):

$$EF' = \min(\max(EF + \Delta_R, 1.1), 2.5) \quad (2)$$

Where Δ_R values are:

TABLE I. Δ_R VALUES

Response	Δ_R
Wrong	-0.25
Hard	-0.20
Normal	0
Easy	+0.15

The learning phase prioritizes newly added and lapsed cards, with time-based intervals of 5, 10, and 25 minutes. Correct answers allow cards to progress to the regular review cycle, while incorrect answers reset the interval to 5 minutes, reinforcing learning through repeated engagement. To encourage consistent participation, a notification system reminds students to complete their due and learning-phase questions daily.

The system utilizes a hybrid Convolutional Neural Network (CNN) architecture with two primary components: a Facial Recognition Model and an Engagement Detection Model. The Facial Recognition Model, trained on the FER-2013 dataset, tracks facial expressions, eye gaze, and head movements, which are essential for assessing engagement. The FER-2013 dataset includes 35,887 grayscale images at a 48x48 pixel resolution, covering seven emotional categories: anger, disgust, fear, happiness, sadness, surprise, and neutral, with diversity across age, ethnicity, and gender for robust emotion detection. To detect head movement, the system leverages dlib for facial landmark detection and OpenCV for frame processing and head pose estimation. Integrated with the Adult ADHD Self-Report Scale (ASRS), a Behavior Classification Function categorizes behaviors into the following: ADHD symptoms with both emotional and head movement alterations, ADHD symptoms with emotional alterations, ADHD symptoms with head movement alterations, ADHD symptoms detected consider monitoring, no ADHD symptoms with both emotional and head movement alterations, no ADHD symptoms with emotional

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alterations, no ADHD symptoms with head movement alterations, no ADHD symptoms.

Engagement data from the CNN system triggers personalized break recommendations to optimize learning, evaluated through metrics such as precision, recall, and F1-score, enabling continuous refinement of the tool for students with inattentive ADHD.

Inspired by prior research, the tool predicts academic performance, offers personalized feedback, analyzes learning patterns, and provides behavioral insights for academic improvement. It dynamically assigns tasks based on individual characteristics such as focus levels and quiz scores. Data is imported into Python for analysis and model building, with preprocessing including scaling via StandardScaler. The dataset is divided using the train-test split, monitoring metrics such as loss and accuracy during training with categorical cross entropy as the loss function. Model performance is evaluated on training and validation datasets to minimize overfitting and ensure generalization [21].

The Adaptive Learning Tool provides real-time feedback, dynamically assigning tasks based on individual strengths and weaknesses. Matplotlib visualizations depict training dynamics, aiding in model interpretation and identifying areas for improvement. This iterative process allows for continuous refinement based on participant feedback and ongoing evaluation of model performance, and this methodology enhances task management, feedback mechanisms with predictions, and a full analysis dashboard for the student to improve learning outcomes for students with inattentive ADHD. This component uses an adaptive algorithm to enhance educational outcomes for ADHD students by integrating data from quiz scores, focus levels, and study time. A regression model predicts potential exam scores based on real-time data, which informs personalized task generation. Tasks are dynamically assigned based on performance type and weak areas, with feedback provided through OpenAI. The dashboard visualizes student progress, supporting ongoing improvements in learning outcomes**.**

IV. Results and Discussion

The study aimed at improving educational outcomes for students with ADHD through the development and implementation of an adaptive learning system yielded promising results across its four major components. These components, Content Chunking and Simplification with Interactive Content, an Adaptive Learning Tool for Undergraduate Students, an ADHD Identification and Support Tool, and an Adaptive Quizzing System, each demonstrated significant positive impacts on the learning experiences and academic performances of students with ADHD.

TABLE II. Performance Matric For NLP Models

Metric	NLP Transformer Models	
	BERT (Extractive)	***T5 (Abstractive)***
ROUGE Score	0.55	0.65
Inference Time (Seconds)	0.5	1.2
Readability	6.5/10	8.0/10

Table II compares BERT (extractive summarization) and T5 (abstractive summarization) across three key metrics: ROUGE score, inference time, and readability. The ROUGE score, which ranges from 0 to 1 and measures the overlap between generated and reference summaries, shows T5 performing better (0.65) than BERT (0.55) due to T5's ability to generate more comprehensive summaries. Inference time reflects the speed of summary generation, with BERT being faster (0.5 seconds) as it selects existing text, while T5 takes longer (1.2 seconds) due to sentence generation. Readability, which evaluates how easy the summaries are to understand, favors T5 with a score of 8.0/10 compared to BERT's 6.5/10, as T5's summaries tend to be more natural and coherent. Given its superior performance in ROUGE score and readability, T5 is selected as the preferred model for summarization because the generated content should be as coherent as possible despite its slightly higher computational cost.

The application of Content Chunking and Simplification through NLP technologies markedly improved learning outcomes for students with ADHD. Feedback from users indicated that restructured materials, which emphasized clear key points and manageable chunks, were significantly more user friendly and engaging.

The use of interactive mind maps greatly enhanced students' comprehension and memory retention. Quantitative data supported these observations, showing notable academic performance gains characterized by higher test scores and reduced study time. Personalized content features such as adjustable presentation formats, font sizes, and color schemes led to increased student engagement and satisfaction. Additionally, observations from psychiatrists and therapists revealed improvements in attention spans and cognitive performance in students utilizing these materials. These findings align with existing research that highlights the benefits of structured and personalized educational strategies for students with ADHD. Future studies should explore the broader applications, long-term effects, and potential of advanced NLP technologies to further tailor educational content, thereby fostering a more inclusive and effective learning environment for ADHD students.

The Adaptive Learning Tool for Undergraduate Students with inattentive ADHD showed very encouraging results. The performance analysis model demonstrated a high accuracy rate of 70% for predicting and analyzing student performance.

TABLE III. Performance Matric For The Performance Analysis Model

Metric	Score
F1 Score	82.3
Predictive Accuracy	81.7
Detection rate (Recall)	80.5
Memory Retention Improvement	40.0

Table III summarizes the performance of the performance analysis model designed to enhance memory retention in students with ADHD. The results indicate a significant improvement in memory retention and strong performance across F1, precision, and recall metrics. This tool had a significant impact on academic performance, with users experiencing a 12% increase in grades. Students reported improved focus and reduced distractions, which translated

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into better exam scores and overall academic performance. Student feedback was overwhelmingly positive, with 85% appreciating the relevance and ease of handling of the assigned tasks.

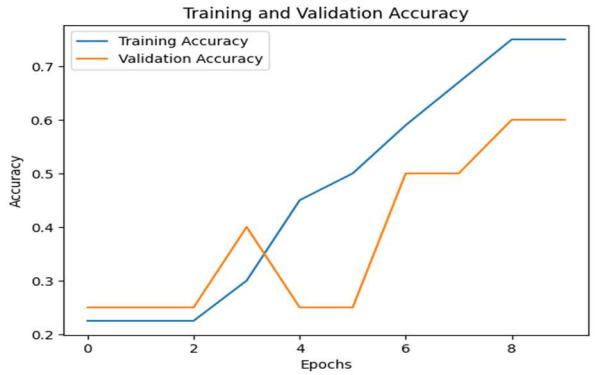


Fig. 2. Accuracy Graph for Performance Analysis Model

Educators also reported enhanced monitoring of student progress and the timely interventions that this tool facilitated. The success of this adaptive learning tool underscores its potential for broader application, showcasing how adaptive learning technologies can elevate the educational experience for inattentive students with ADHD, making it more inclusive and effective. These results contribute to the broader field of technology enhanced learning by providing a scalable solution tailored to various educational contexts.

In developing the ADHD Identification and Support Tool, significant strides were made in identifying ADHD symptoms. And providing personalized support to students. By implementing the ASRS v1.1 questionnaire for behavioral analysis and utilizing advanced techniques for emotion detection and head movement tracking, the system could monitor real-time behavioral and emotional data.

TABLE IV. COMPARATIVE PERFORMANCE OF DL ARCHITECTURES

DL Architectures	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	Loss Value
VGG16	73.00	72.00	72.50	72.75	0.55
ResNet50	76.92	75.00	78.00	76.50	0.50
Custom CNN	85.90	84.00	86.00	84.75	0.39

As can be seen in Table IV, the accuracy curves below indicate that the Custom CNN model learned effectively throughout the 50 epochs, with training accuracy steadily increasing and validation accuracy stabilizing around 62.40%. The classification tree effectively categorized ADHD symptoms considering emotional and head movement alterations. The results indicated that the tool could accurately detect ADHD related behaviors and recommend appropriate breaks and learning adaptations. Real time feedback and personalized interventions significantly enhanced student engagement and focus, demonstrating the tool's potential to improve educational outcomes for students with inattentive ADHD. Future efforts will focus on refining the algorithms and expanding the dataset to enhance accuracy and broaden the tool's application. According to Fig. 3, Despite the strong performance, the loss curves suggest slight overfitting

beginning after 40 epochs, as indicated by the divergence between training and validation loss.

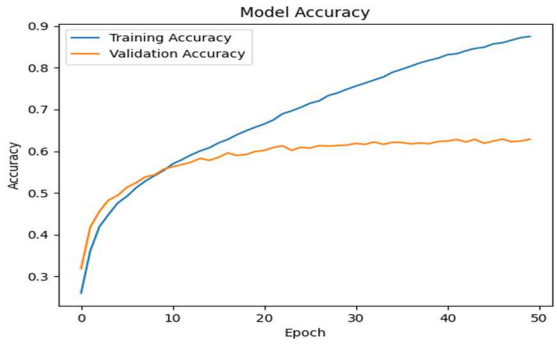


Fig. 3. Training and Validation Accuracy

The Adaptive Quizzing Tool significantly enhanced memory retention and engagement for students with ADHD through a combination of advanced natural language processing and a modified SuperMemo-2 (SM-2) algorithm. Initially, key phrases were extracted using GPT-3, which also generated distractors to form multiple-choice questions (MCQs). These questions were refined and generated using a fine-tuned T5 model, ensuring high relevance and accuracy.

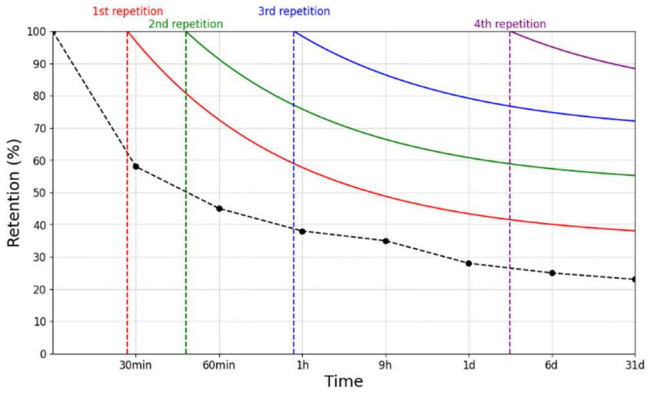


Fig. 4. Alteration of the Forgetting Curve

The SM-2 algorithm was adapted to reduce recall intervals, specifically tailored to the cognitive needs of ADHD students. This modification enabled the tool to achieve retention rates comparable to those seen with the traditional SM-2 algorithm, which has demonstrated a 25.3% improvement in retention [22]. This tailored approach suggests that long-term retention can be enhanced, potentially reaching up to 90%, aligning with the observed benefits in traditional spaced repetition systems. As shown in Fig. 4, the alteration of the forgetting curve illustrates the adjustment of the forgetting curve, emphasizing how the optimized repetition intervals reduce memory decay. The model's training performance is shown in Fig. 5, where the consistent decrease in training and validation loss underscores the model's strong generalization capabilities.

Evaluation metrics presented in Table V demonstrate the tool's effectiveness, with precision, recall, and F1 scores exceeding 85%, validating its suitability for adaptive learning environments.

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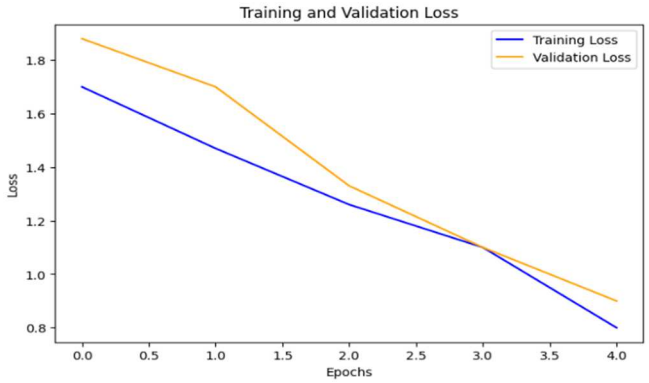


Fig. 5. Training and Validation Loss Curves

TABLE V. TRAINING AND VALIDATION LOSS CURVES

Metric	Score (%)
Precision	87.5
Recall	85.3
F1 Score	86.4

Overall, the results from this study indicate that the developed adaptive learning tool for students with inattentive ADHD has shown substantial positive impacts on academic performance, memory retention, and overall engagement, highlighting the importance of tailored educational solutions for neurodiverse learners.

V. CONCLUSION & FUTURE WORK

The study presented an adaptive learning tool with four main components: personalized ADHD identifier and break recommendation, adaptive quizzing, content chunking and simplification, and learning potential maximization. The tool's goal is to improve educational outcomes for students with inattentive ADHD. By using natural language processing (NLP) technologies, educational materials were simplified and made easier to grasp, improving exam results and understanding. By offering regular tailored tests, the Adaptive Quizzing System which makes use of a modified spaced repetition algorithm improved memory retention and engagement. The Personalized ADHD Identifier and Break Recommendation System enhanced focus and cognitive function by recommending beneficial breaks based on an analysis of real-time webcam data using hybrid CNN models. By using machine learning to provide individualized feedback and assign assignments in an adaptable manner, the Learning Potential Maximization tool improved student performance. The results highlight how adaptive learning technologies can be used to design inclusive classrooms that are specifically designed to meet the needs of students with ADHD. By customizing learning experiences and providing on demand support, these technologies can greatly improve student engagement, memory retention, and academic performance. Moving forward, future research aims to conduct extensive longitudinal studies to assess the tool's long-term efficacy, focusing on sustained engagement, enhanced focus, and improved academic outcomes. With flexible architecture, this tool is crafted for scalability, enabling integration across varied educational settings, from schools to universities. This adaptability not only broadens its reach but also underscores its potential as a transformative approach to personalized learning. Continuous development and rigorous testing will ensure continuous improvement, refining its adaptability and maximizing its educational impact to support a wide range of learners with ADHD and similar cognitive challenges.

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